

Classification and types of mineral deposits

A mineral deposit (or an ore deposit) may be defined as a rock body that contains one or more elements (or minerals) sufficiently above the average crustal abundance to have potential economic value.

The mineral deposit can be broadly categorised into

- 1) metallic ore mineral deposit from which one or more metal can be extracted and
- 2) non-metallic or industrial mineral deposits.

Geologists classify mineral deposits in many different ways, according to the:

- 1) Commodity being mined
- 2) Tectonic setting in which the deposit occurs
- 3) Geological setting of the mineral deposit
- 4) Genetic model for the origin of the ore deposit

Classification of metallic and non-metallic minerals:

Metallic Minerals

1. Ferrous Metals (Fe, Ti, Cr, Mn etc)
2. Non-Ferrous Metals (Cu, Pb, Zn, Sb, Ni etc)
3. Noble Metals (Au, Ag, Pt, Os, Ir etc)
4. Light Metals (Al, Li, Be, Mg etc)
5. Rare earths and dispersed Metals (La, Ce to Lu) and (Sc, Ga, Rb, Cd etc)
6. Rare and Scarce Metals (W, Mo, Sn, Co, Hg, Bi, Zr, Cs, Nb, Ta etc)

Non Metallic Materials

1. Fossil Fuels (Coal, Petroleum and natural gas)
2. Structural and building material (Granite, Khondalite, Dolerite, Slate, Gypsum, etc)
3. Ceramic materials (Clays, Feldspar, Bauxite, Magnesite, Barite etc)
4. Metallurgical and Refractory materials (Fluorspar, Graphite, Magnesite, dolomite, Bauxite, Zircon etc)
5. Industrial and manufacturing materials (Asbestos, Mica, Talc, Barite etc)
6. Chemical materials (Salt, Borates, Ca, Mg, Cl₂, Bromine, Nitrates etc)
7. Abrasives and abrasive materials (Corundum, Emery, Garnet etc)

Two groups of mineral deposits may be established, syngenetic and epigenetic depending on their time-relationship to the rocks enclosing or associated with them.

Syngenetic

A syngenetic mineral deposit is a deposit that is formed at the same time as the rock that it encloses. Magmatic deposits are syngenetic in that the ore minerals crystallize from the same liquid that produces the silicate minerals which form the bulk of the intrusive – they crystallize more or less simultaneously as the melt cools.

Process: Magmatic intrusion and later transported by hydrothermal fluids, VMS, Porphyry, SEDEX

Epigenetic

If a mineral deposit formed much later than the rocks which enclose it, it is said to be epigenetic. An example is a vein. The first step in the formation of a vein is the fracturing or breaking of rock along a fault zone, at a depth ranging from surface to several kilometers below surface. The rock must be solid (lithified) and brittle, creating open spaces when it breaks. Hydrothermal solutions pass along the fault zone and deposit or precipitate the ore and gangue minerals within.

Genetic Classification of Mineral Deposits

The most commonly used scheme is the genetic classification scheme. These deposits include:

A. Magmatic deposits:

1. Early Magmatic:
 - a. Dissemination: Disseminated crystallization without concentration: eg. Diamond, Panna, M.P.
 - b. Segregation: Crystallisation, differentiation and accumulation: eg. Chromite, Singhbhum Dt., Jharkhand.
2. Late Magmatic:
 - a. Gravitative liquid accumulation
 - i. Residual liquid segregation: Crystallisation, differentiation and residual magma accumulation: eg. Chromites & Vanadiferous, Titanomagnetite of Mayurbhanj Dt. Orissa.
 - ii. Residual liquid injection: Crystallisation, differentiation. Residual magma accumulation with filter pressing and/or injection: eg. Pegmatites hosted mica deposits, Giridih, Jharkhand.
 - b. Immiscible Liquid accumulation
Immiscible liquid segregation: Immiscible liquid segregation and accumulation: eg. 1. Magmatic ilmenite deposits; 2. Ni-Sulphide deposits.

Orthomagmatic deposits are those that form from primary magmatic processes (i.e. magmas). They are hosted in the igneous rocks in which they have formed. Most of the world's nickel, chromium and platinum-group elements are derived from these deposits. The largest deposits of platinum-group elements and chromium come from the 2055 million year old Bushveld Complex of northern South Africa. **Orthomagmatic deposits include: chromium, titanium, iron, nickel, copper, platinum-group elements, diamonds**

Pneumatolytic and pegmatitic deposits are formed from volatile-rich (i.e. rich in water, fluorine, boron) high-temperature fluids emanating from igneous intrusions. Some of these deposits occur as pipe-like bodies or breccia pipes. These are important sources for: tin, rare-earth elements, tantalum, beryllium, lithium, molybdenum, tungsten

B. Hydrothermal Deposits

Hydrothermal deposits cover a wide range of different deposits types but all form from hot circulating water-rich fluids. These include the two main types of gold deposits - epithermal and lode gold deposits along with replacement deposits in calcareous sequences (Mississippi Valley deposits), base-metal vein deposits and replacement skarn deposits. The term hydrothermal deposits have formed from mineralised hot aqueous saline electrolyte solutions which have sources either from meteoric, magmatic, metamorphic or sedimentary trapped solutions which scavenge the metals from the Earth's crust and localize the metals in the favorable loci. They are further classified as hypothermal, mesothermal, epithermal, telethermal and xenothermal categories based on the temperature and pressure conditions. The main metallic commodities found in this type are gold, copper, lead, Zinc, etc.

C. Metamorphic Ore Deposits

The process of metamorphism gave rise to some of the major deposits of manganese, copper, iron, lead, zinc as well as some industrial minerals such as asbestos, magnesite, talc and graphite.

D. Sedimentary Ore Deposits

Coal, Petroleum and natural gas and many metallic ore deposits like copper, iron and uranium, stratified Pb, Zn deposits occur within or part of the sedimentary strata. Most of the world's iron and manganese are derived from deposits of this type. These deposits are very large in size (thousands of millions of tons) and are usually mined by open-cut methods. The iron and manganese ores occur as distinct sedimentary layers alternating with iron and manganese-poor sedimentary layers - the whole mass is sometimes mined. Iron ores of this type are commonly termed the banded iron formation or simply abbreviated as BIF.

The most common iron ore minerals are hematite, lepidocrocite and goethite while the most common manganese ore minerals are braunite, manganite and hausmannite. These deposits are believed to have formed in shallow marine basins and are confined to Proterozoic rock sequences

i) Placer deposits

These are made of alluvial, colluvial and eluvial material, which contain economic quantities of some valuable minerals:

Alluvial: Detrital material which is transported by a river and usually deposited along the river's pathway, either in the riverbed itself or on its floodplain.

Colluvial: Weathered material transported by gravity action such as on scree slopes.

Eluvial: Weathered material still at or near its point of formation.

The most common placer deposits are those of gold, platinum, group minerals, gemstones, tin, rutile, monazite and zircon.

ii) Evaporite deposits

An evaporite is a sediment that forms through the evaporation of saline water. The most common evaporite deposits are salts (most commonly sylvite and halite), gypsum, and nitrates. Most evaporites are derived from bodies of sea-water, but under special conditions, inland lakes may also give rise to evaporite deposits, particularly in regions of low rainfall and high temperature. The original character of most evaporite deposits has been destroyed by replacement through circulating fluids.

iii) Phosphatic deposits

Phosphorite is a commonly used term for lithified phosphate rock. The island of Nauru in the Pacific is one of the world's largest deposits of phosphorite and has been mined since at least the 1950s. The mineralogy of phosphate deposits is very complex. They usually consist of fine-grained mixtures of various calcium phosphates with the most common mineral being apatite. Collophane is an amorphous calcium phosphate that is also commonly found in phosphate deposits.

iv) Residual deposits

Residual deposits are formed in tropical regions. During the wet season, intense leaching of the rock occurs. Then, during the dry season, the solution containing the leached ions is drawn towards the surface by capillary action where it evaporates leaving behind salts that are washed away in the next wet season. Eventually, the whole zone down to the base of the water table is leached of relatively mobile ions such as sodium, potassium, calcium and magnesium. Basic and ultrabasic rocks tend to form laterites while granitic rocks lead to the development of bauxite